



14.1.1 Conditional Probability: NIELIT Scientific Assistant A 2020 November: 99

A bag contains 10 white balls and 5 blue balls. A ball is drawn from the bag and its color is noted. This ball is put back in the bag along with 3 more balls of the same color. A ball is drawn again from the bag at random. The probability that the first ball drawn is blue, given that the second ball drawn is blue, is:

- A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{8}{9}$ D. $\frac{4}{9}$

nielit-sta-2020 probability conditional-probability

Answer key

14.2.1 Counting: NIELIT 2021 Dec Scientist A - Section A: 25



Out of 13 applicants for a job there are 5 women and 8 men. Two persons are to be selected for the job. Find the probability that at least one of the selected persons will be a women.

- A. $\frac{25}{39}$ B. $\frac{10}{21}$ C. $\frac{14}{27}$ D. $\frac{12}{51}$

nielit2021dec-scientista probability counting

Answer key

14.2.2 Counting: NIELIT Scientist B 2020 November: 38



In a group of 24 members, each member drinks either tea or coffee or both. If 15 of them drink tea and 18 drink coffee, find the probability that a person selected from the group drinks both tea and coffee.

- A. $\frac{1}{8}$ B. $\frac{3}{8}$ C. $\frac{5}{24}$ D. None of the options

nielit-scb-2020 probability counting analytical-aptitude

Answer key

14.3.1 Expectation: NIELIT 2021 Dec Scientist B - Section B: 52



Let the random variable X has a mixed distributions with probability $P(X=0) = \alpha$, and the density function.

$$f_x(x) = \begin{cases} \beta x^2(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If the expectation of X is α , then the value of $4\alpha + \beta$ is equal to :

- A. $9/2$ B. 6 C. 9 D. $5/2$

nielit-2021-it-dec-scientistb probability expectation random-variable nielit2021dec-scientistb

14.3.2 Expectation: NIELIT 2021 Dec Scientist B - Section B: 71



Let the random variable X has a mixed distributions with probability $P(X = 0) = \alpha$, and the density function.

$$f_x(x) = \begin{cases} \beta x^2(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If the expectation of X is α , then the value of $4\alpha + \beta$ is equal to :

- A. $9/2$ B. 6 C. 9 D. $5/2$

nielit2021dec-scientistb probability random-variable expectation

14.3.3 Expectation: NIELIT Scientist B 2020 November: 29



A player rolls a die and receives the same number of rupees as the number of dots on the face that turns up. What should the player pay for each roll if he wants to make a profit of one rupee per throw of the die in the long run?

- A. ₹ 2.50 B. ₹ 2 C. ₹ 3.50 D. ₹ 4

nielit-scb-2020 probability expectation quantitative-aptitude

Answer key

14.4 Independent Events (1)

14.4.1 Independent Events: NIELIT 2021 Dec Scientist A - Section B: 79



Let X be uniform random variable on $[0, 4]$ and Y be uniform random variable on $[0, 1]$. If X and Y are independent, then $P(\max\{X, Y\} > 3)$ is equal to :

- A. $1/4$ B. $1/2$ C. $1/8$ D. 1

nielit2021dec-scientista probability random-variable independent-events

Answer key

14.5 Normal Distribution (1)

14.5.1 Normal Distribution: NIELIT 2018-24



If $f(x) = k \exp\{-(9x^2 - 12x + 13)\}$, is a p, d, f of a normal distribution (k , being a constant), the mean and standard deviation of the distribution:

- A. $\mu = \frac{2}{3}, \sigma = \frac{1}{3\sqrt{2}}$ B. $\mu = 2, \sigma = \frac{1}{\sqrt{2}}$

C. $\mu = \frac{1}{3}, \sigma = \frac{1}{3\sqrt{2}}$

D. $\mu = \frac{2}{3}, \sigma = \frac{1}{\sqrt{3}}$

nielit-2018 engineering-mathematics probability normal-distribution

Answer key

14.6

Probability (10)

14.6.1 Probability: NIELIT 2016 MAR Scientist B - Section B: 16



A box contains 10 screws, 3 of which are defective. Two screws are drawn at random with replacement. The probability that none of two screws is defective will be

- A. 100% B. 50% C. 49% D. None of these.

nielit2016mar-scientistb engineering-mathematics probability

Answer key

14.6.2 Probability: NIELIT 2016 MAR Scientist C - Section B: 4



If A and B are two related events, and $P(A | B)$ represents the conditional probability, Bayes' theorem states that

- A. $P(A | B) = \frac{P(A)}{P(B)} P(B | A)$ B. $P(A | B) = P(A)P(B)P(B | A)$
 C. $P(A | B) = \frac{P(A)}{P(B)}$ D. $P(A | B) = P(A) + P(B)$

nielit2016mar-scientistc engineering-mathematics probability

Answer key

14.6.3 Probability: NIELIT 2017 DEC Scientific Assistant A - Section B: 41



If a random coin is tossed 11 times, then what is the probability that for 7th toss head appears exactly 4 times?

- A. $5/32$ B. $15/128$ C. $35/128$ D. None of the options

nielit2017dec-assistanta engineering-mathematics probability

Answer key

14.6.4 Probability: NIELIT 2017 DEC Scientific Assistant A - Section B: 44



If X, Y and Z are three exhaustive and mutually exclusive events related with any experiment and the $P(X) = 0.5P(Y)$ and $P(Z) = 0.3P(Y)$. Then $P(Y) =$ _____.

- A. 0.54 B. 0.66 C. 0.33 D. 0.44

nielit2017dec-assistanta engineering-mathematics probability

Answer key

14.6.5 Probability: NIELIT 2017 July Scientist B (IT) - Section B: 23



The probability that top and bottom cards of a randomly shuffled deck are both aces is:

- A. $\frac{4}{52} \times \frac{4}{52}$
C. $\frac{4}{52} \times \frac{3}{51}$

- B. $\frac{4}{52} \times \frac{3}{52}$
D. $\frac{4}{52} \times \frac{4}{51}$

nielit2017july-scientistb-it engineering-mathematics probability

Answer key

14.6.6 Probability: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 16



If P is risk probability, L is loss, then Risk Exposure (RE) is computed as.

- A. $RE = P/L$
C. $RE = P * L$

- B. $RE = P + L$
D. $RE = 2 * P * L$

nielit2017oct-assistanta-cs engineering-mathematics probability

Answer key

14.6.7 Probability: NIELIT 2018-26



If $P(A \cap B) = \frac{1}{2}$, $P(\bar{A} \cap \bar{B}) = \frac{1}{2}$ and $2P(A) = P(B) = p$, then the value of p is given by:

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

nielit-2018 engineering-mathematics probability

14.6.8 Probability: NIELIT 2021 Dec Scientist A - Section A: 34



First bag contains 5 white and 4 black balls. Second bag contains 7 white and 9 black balls. A ball is transferred from the first bag to the second bag and then a ball is drawn from the second bag. Find the probability that the ball drawn is white.

A. $\frac{7}{18}$

B. $\frac{5}{9}$

C. $\frac{4}{9}$

D. $\frac{11}{18}$

nielit2021dec-scientista probability

Answer key

14.6.9 Probability: NIELIT 2021 Dec Scientist A - Section B: 108



The covariance function of a band limited white noise is :

A. A Dirac delta function

B. An exponentially decreasing function

C. A sinc function

D. A sinc² function

nielit2021dec-scientista probability

Answer key

14.6.10 Probability: NIELIT 2022 April Scientist B | Section B | Question: 114



For the probability density function $f(x) = cx^2(1 - x)$, $0 < x < 1$, the value of constant c and mean of the distribution are :

- A. 6,0.4 B. 8,0.5 C. 12,0.6 D. 8,0.6

nielit2022apr-scientistb probability

Answer key

14.7

Random Variable (1)

14.7.1 Random Variable: NIELIT 2018-14



A continuous random variable x is distributed over the interval $[0, 2]$ with probability density function $f(x) = ax^2 + bx$, where a and b are constants. If the mean of the distribution is $\frac{1}{2}$. Find the values of the constants a and b .

- A. $a = 2, b = -\frac{13}{6}$ B. $a = -\frac{15}{8}, b = 3$
C. $a = -\frac{29}{6}, b = 2$ D. $a = 3, b = -\frac{7}{2}$

nielit-2018 engineering-mathematics probability random-variable

Answer key

14.8

Statistics (1)

14.8.1 Statistics: NIELIT 2016 MAR Scientist B - Section B: 17



Following marks are obtained by the students in a test:

81, 72, 90, 90, 86, 85, 92, 70, 71, 83, 89, 95, 85, 79, 62. Range of the marks is

- A. 9 B. 17 C. 27 D. 33

nielit2016mar-scientistb statistics

Answer key

14.9

Variance (1)

14.9.1 Variance: NIELIT Scientist B 2020 November: 112



Let $X_1, \dots, X_{50}$ be independent random variables following $N(0, 1)$ distribution.

Let $\sum_{i=1}^{50} X_i^2$, and $E(Y) = a$ and $Var(Y) = b$. Then, the ordered pair (a, b) is:

- A. (50, 100) B. (50, 50) C. (25, 50) D. (25, 100)

nielit-scb-2020 normal-distribution probability random-variable expectation variance

Answer Keys

14.1.1	D
14.3.3	A
14.6.3	A
14.6.8	C
14.9.1	A

14.2.1	A
14.4.1	A
14.6.4	A
14.6.9	C

14.2.2	B
14.5.1	A
14.6.5	C
14.6.10	C

14.3.1	C
14.6.1	C
14.6.6	A
14.7.1	B

14.3.2	C
14.6.2	A
14.6.7	D
14.8.1	D